Chapter 2: Basic Topology

Section: Perfect Sets

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

The Main Theorem

The Cantor Set

Why Perfect Sets Matter

- A perfect set is closed with *no isolated points*

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- Every nonempty perfect set in $\mathbb{R}^k$ is uncountable

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- A perfect set is closed with *no isolated points*
- Every nonempty perfect set in $\mathbb{R}^k$ is uncountable
- The Cantor set: perfect, uncountable, contains no interval

The Main Theorem

Recall: Definition 2.18(h) — Perfect Set

Definition (Perfect Set)

A set E in a metric space X is perfect if

(1) E is closed, and

(2) every point of E is a *limit point* of E.

Equivalent form: E is perfect $\iff$ E has no isolated points and contains all its limit points.

Theorem 2.43: Perfect Sets Are Uncountable

Theorem 2.43

Let P be a nonempty perfect set in $\mathbb{R}^k$.
Then P is uncountable.

Contrast: A countable set can still be closed — e.g. $\mathbb{Z}$ or $\{1/n\} \cup \{0\}$.

What fails for these sets? They have *isolated points*.

Proof Strategy: Theorem 2.43

Approach: Proof by contradiction, using compactness.

Setup: Assume P is countable, so

$$P = \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots\}.$$

Plan: Build a shrinking sequence of neighborhoods $V_1 \supset V_2 \supset \cdots$ such that

- each V_n meets P,
- $\overline{V}_{n+1}$ excludes $\mathbf{x}_n$ — so no listed point lies in every $\overline{V}_n$.

Then nested *compact* sets $K_n = \overline{V}_n \cap P$ must share a point of P — contradiction.

Proof — Step 1: The Base Case

Step 1: Choose V_1

Let V_1 be *any* neighborhood of $\mathbf{x}_1$:

$$V_1 = \{\mathbf{y} \in \mathbb{R}^k : |\mathbf{y} - \mathbf{x}_1| < r\}.$$

Its closure is the closed ball

$$\overline{V}_1 = \{\mathbf{y} \in \mathbb{R}^k : |\mathbf{y} - \mathbf{x}_1| \leq r\}.$$

Since $\mathbf{x}_1 \in P$, we have $V_1 \cap P \neq \emptyset$.

Proof — Step 2: The Inductive Step

Step 2: Build V_{n+1} from V_n

Inductive hypothesis: V_n is a neighborhood with $V_n \cap P \neq \emptyset$.

We want to construct V_{n+1} satisfying:

(i) $\overline{V}_{n+1} \subset V_n$

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(ii) $\mathbf{x}_n \notin \overline{V}_{n+1}$ (kick out the n-th listed point)

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(ii) $\mathbf{x}_n \notin \overline{V}_{n+1}$ (kick out the n-th listed point)

(iii) $V_{n+1} \cap P \neq \emptyset$ (propagate the hypothesis)

Proof — Step 2: The Inductive Step

Why this is possible

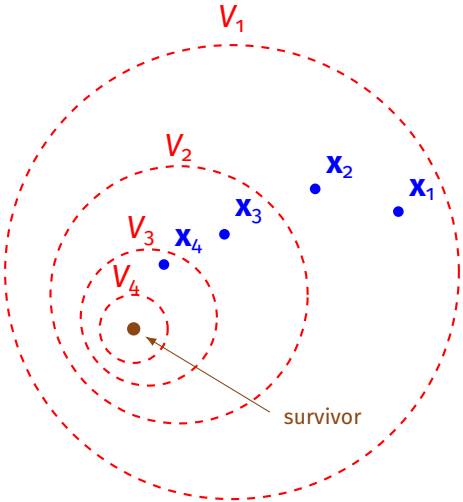
Pick any $\mathbf{p} \in V_n \cap P$. Since P is perfect, $\mathbf{p}$ is a limit point of P, so arbitrarily small neighborhoods of $\mathbf{p}$ contain many points of P.

Choose $\mathbf{q} \in P$ with $\mathbf{q} \neq \mathbf{x}_n$ arbitrarily close to $\mathbf{p}$.
Take V_{n+1} a tiny ball around $\mathbf{q}$ whose closure:

- stays inside V_n [gives (i)]
- misses $\mathbf{x}_n$ [gives (ii)]

Since $\mathbf{q} \in V_{n+1} \cap P$, we have (iii).

Visualizing the Shrinking Neighborhoods



Proof — Step 3: Compact Pieces K_n

Define

$$K_n = \overline{V}_n \cap P$$

Each K_n is compact:

- $\overline{V}_n$ is closed and bounded in $\mathbb{R}^k \implies$ compact (Heine–Borel).
- P is closed.
- $K_n = \overline{V}_n \cap P$ is a closed subset of a compact set $\implies$ compact (Thm 2.35).

Proof — Step 4: Reaching a Contradiction

Properties of $\{K_n\}$

(a) Each K_n is nonempty: by (iii), $V_n \cap P \neq \emptyset$, so $K_n \supset V_n \cap P \neq \emptyset$.

Proof — Step 4: Reaching a Contradiction

Properties of $\{K_n\}$

(a) Each K_n is nonempty: by (iii), $V_n \cap P \neq \emptyset$, so $K_n \supset V_n \cap P \neq \emptyset$.

(b) $K_n \supset K_{n+1}$: by (i), $\overline{V}_{n+1} \subset V_n \subset \overline{V}_n$.

Proof — Step 4: Reaching a Contradiction

Properties of $\{K_n\}$

(a) Each K_n is nonempty

(b) $K_n \supset K_{n+1}$

(c) $\mathbf{x}_n \notin K_{n+1}$: by (ii), $\mathbf{x}_n \notin \overline{V}_{n+1}$, so $\mathbf{x}_n \notin K_{n+1}$.

Therefore no listed point $\mathbf{x}_n$ lies in $\bigcap_{m=1}^{\infty} K_m$, so

$$\bigcap_{m=1}^{\infty} K_m \cap P = \emptyset.$$

But $K_m \subset P$ for all m, so $\bigcap_{m=1}^{\infty} K_m = \emptyset$.

Proof — Conclusion

We have a sequence of nonempty nested compact sets

$$K_1 \supset K_2 \supset K_3 \supset \cdots$$

with $\bigcap_{n=1}^{\infty} K_n = \emptyset$.

But the *Corollary to Theorem 2.36* says

$$\bigcap_{n=1}^{\infty} K_n \neq \emptyset.$$

Contradiction! $\boxed{P \text{ cannot be countable.}}$

Corollary: $\mathbb{R}$ is Uncountable

Corollary

Every interval $[a, b]$ with $a < b$ is uncountable.
In particular, $\mathbb{R}$ is uncountable.

Why: $[a, b] \subset \mathbb{R}$ is

- closed,
- every point is a limit point (interior points approached from both sides; endpoints from inside),

so $[a, b]$ is perfect. Apply Theorem 2.43.

The Cantor Set

2.44 — The Cantor Set

Goal: Exhibit a perfect set in $\mathbb{R}^1$ that contains no segment.

Why is this surprising?

- The obvious perfect subsets of $\mathbb{R}$ — closed intervals — are "thick".
- Can we have a perfect set that is so "thin" it contains no interval?

Answer: Yes — the Cantor set.

Construction: Removing Middle Thirds

Start with:

$$E_0 = [0, 1].$$

Construction: Removing Middle Thirds

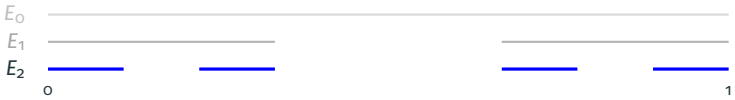
Stage 1: remove the open segment $(\frac{1}{3}, \frac{2}{3})$.

$$E_1 = [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1].$$

Construction: Removing Middle Thirds

Stage 2: remove the middle thirds of each remaining interval.

$$E_2 = [0, \tfrac{1}{9}] \cup [\tfrac{2}{9}, \tfrac{3}{9}] \cup [\tfrac{6}{9}, \tfrac{7}{9}] \cup [\tfrac{8}{9}, 1].$$

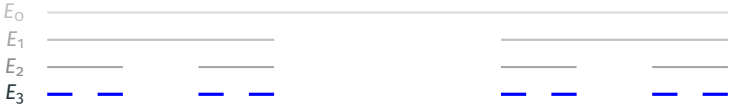


Construction: Removing Middle Thirds

Continuing: we obtain compact sets E_n with

(a) $E_1 \supset E_2 \supset E_3 \supset \cdots$

(b) E_n is the union of 2^n intervals, each of length 3^{-n}.



The Cantor Set

Definition

The Cantor set is

$$P = \bigcap_{n=1}^{\infty} E_n.$$

Basic properties:

- Each E_n is closed and bounded, so compact.
- P is an intersection of compact sets $\implies$ P is compact.
- By the Corollary to Theorem 2.36: $P \neq \emptyset$.

P Contains No Segment

Claim: The Cantor set P contains no segment (α, β).

Removed segments: At stage m, we removed all segments of the form

$$\left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m}\right), \tag{24}$$

where k, m are positive integers. These segments are disjoint from P.

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Removed segments: At stage m, we removed all segments of the form

$$\left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m}\right). \tag{24}$$

Every segment (α, β) contains one of form (24): pick m so that

$$3^{-m} < \frac{\beta - \alpha}{6}.$$

Hence $(\alpha, \beta) \not\subset P$.

$\boxed{P \text{ contains no segment.}}$

P is Perfect: Strategy

P **is closed** — intersection of closed sets.
So we only need to show P has no isolated point.

Plan: For every $x \in P$ and every segment S containing x, find another point of P in S.

Key trick: The endpoints of each stage's intervals are *never removed*, so they all lie in P.

P is Perfect: The Argument

Let $x \in P$ and let S be any segment containing x.

Let I_n be the interval of E_n that contains x.

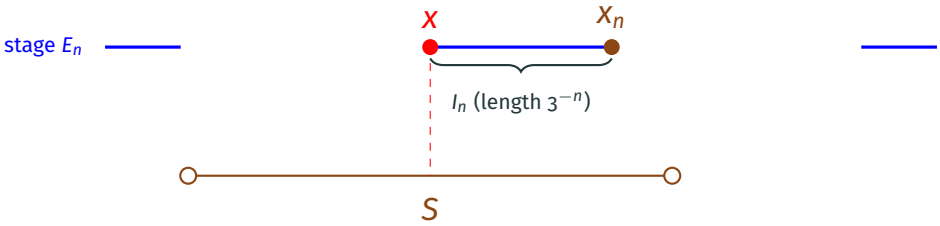
- I_n has length $3^{-n} \to 0$.
- Choose n large enough so $I_n \subset S$.

Let x_n be an endpoint of I_n with $x_n \neq x$.

- $x_n \in P$ (endpoints are never removed).
- $x_n \in I_n \subset S$.

$\Longrightarrow$ x is a limit point of P.

Visualizing Why P is Perfect



Choose n large enough that $I_n \subset S$.

The Cantor Set: A Striking Object

Final observations on 2.44

Combining everything we now know:

- P is perfect and nonempty $\Longrightarrow$ by Theorem 2.43, P is uncountable.
- P contains no segment — topologically "tiny".
- Total length removed: $\frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \cdots = \sum_{n=0}^{\infty} \frac{2^n}{3^{n+1}} = 1.$
- P has measure zero — yet is uncountable (see Chap. 11).

Summary

Key takeaways from this section:

- Theorem 2.43: every nonempty perfect set in $\mathbb{R}^k$ is uncountable.
- Proof technique: nested shrinking neighborhoods + compactness (Cor. 2.36).
- Corollary: every interval $[a, b]$ with $a < b$ is uncountable, so $\mathbb{R}$ is uncountable.
- Cantor set $P = \bigcap_n E_n$ is:
  - compact, nonempty, *perfect*,
  - contains *no segment*,
  - uncountable, measure zero.

Coming up: Connected sets — when is a set "in one piece"?